

# Worksheet IP.4 – Image Processing in Fourier Domain

**Question 1.** A 2D continuous function has maximum frequency components of 60 Hz in the horizontal direction and 45 Hz in the vertical direction. You sample the function at 100 samples per second in the horizontal direction and 80 samples per second in the vertical direction.

1. Determine whether aliasing will occur.
2. If aliasing occurs, suggest a new sampling rate to avoid it.

**Question 2.** You are filtering a  $2048 \times 2048$  image using a  $201 \times 201$  non-separable kernel. Estimate the number of operations required for:

1. Direct spatial filtering
2. Frequency domain filtering using FFT

**Question 3.** You want to convolve two 2D discrete signals of size  $128 \times 128$  and  $64 \times 64$  using Circular DFT. What is the minimum size of zero-padding required to avoid wraparound error?

**Question 4.** Design a Butterworth lowpass filter of order 2 with cutoff frequency  $D_0 = 50$  for a  $256 \times 256$  image. Compute the filter value at point  $(u,v) = (60, 60)$ .